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Studierichting: Informatica
Oefeningenreeks 1C nr. 48.6

$$z^6 = -64$$

$$-64 + 0i = 64 \cos \pi + 64i \sin \pi$$

$$w_0 = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = \sqrt{3} + i$$

$$w_1 = 2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = 2i$$

$$w_2 = 2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = -\sqrt{3} + i$$

$$w_3 = 2 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right) = -\sqrt{3} - i$$

$$w_4 = 2 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) = -2i$$

$$w_5 = 2 \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right) = \sqrt{3} - i$$

References